PCBC H Name _____ _____ /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

PRACTICE C · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

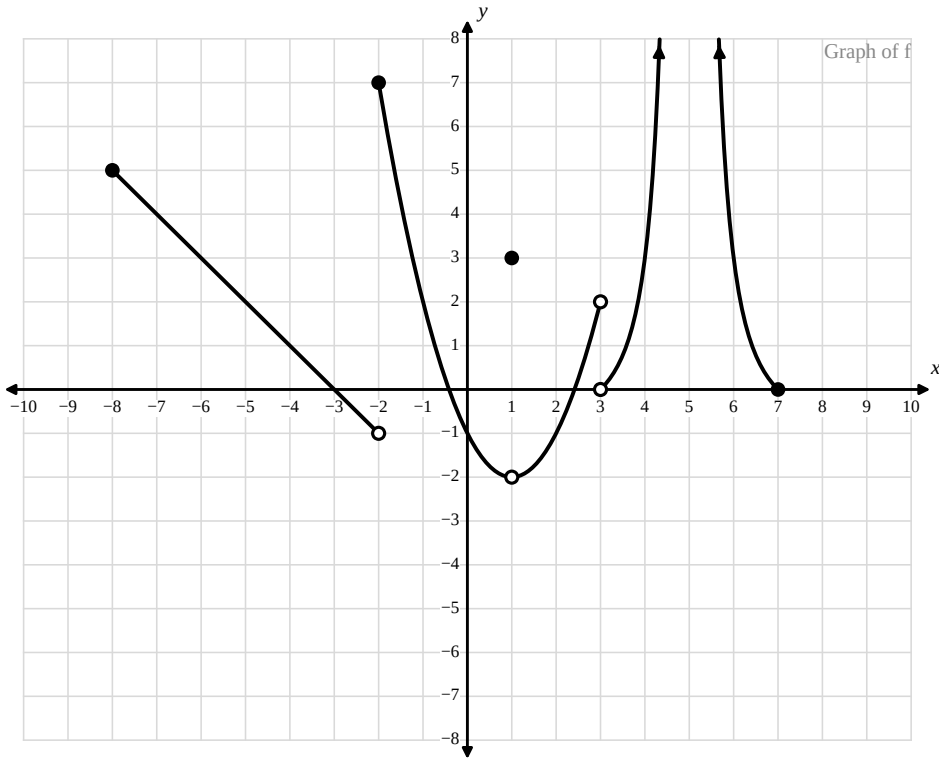
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Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)

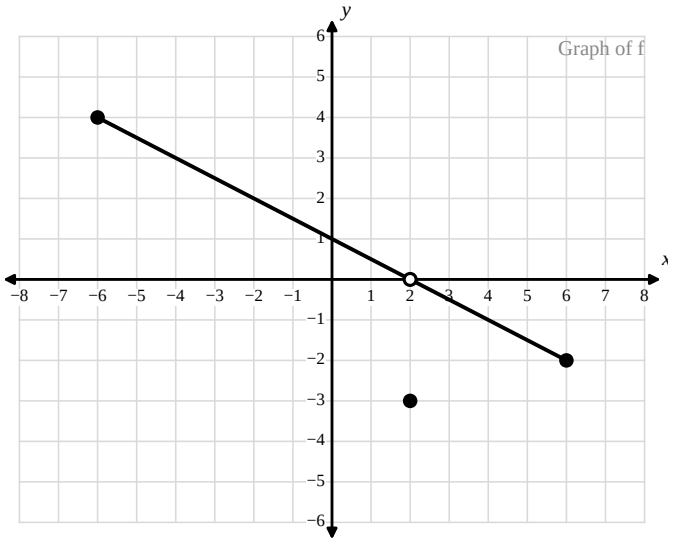


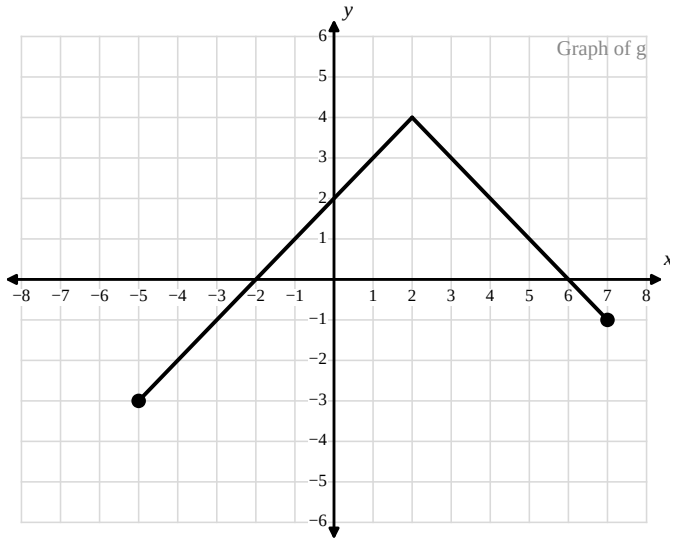
1. $\lim_{x\to 1} f(x) =$	**2.** $\lim_{x\to 3} f(x) =$	**3.** $\lim_{x\to 5} f(x) =$

(2 points)

4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = -1$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim_{x \to 4} \sqrt{[g(x)]^2 - 3f(x)}$

(2 points)

6. Use the table to estimate $\lim_{x \to 2} h(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	1.9	1.99	1.999	$\to 2 \leftarrow$	2.001	2.01	2.1
$h(x)$	11.41	11.9401	11.994001		12.006001	12.0601	12.61

$\lim_{x \to 2^-} h(x) \approx$ _____ $\lim_{x \to 2^+} h(x) \approx$ _____ $\lim_{x \to 2} h(x) \approx$ _____

Evaluate each limit. (2 points each)

7. $\lim_{x \to 1} f(x), \quad f(x) = \begin{cases} \frac{x^2-1}{x-1}, & x < 1 \\ 3 - x^2, & x \geq 1 \end{cases}$

8. $\lim_{x \to \frac{1}{2}} \dfrac{2x^2 + 3x - 2}{4x^2 - 1}$

9. $\lim_{x \to 3} \dfrac{x^2 - 3x}{\sqrt{x+1} - 2}$

10. $\lim_{x \to -1} \dfrac{\frac{1}{x+2} - 1}{x^2 - 1}$

11.
Find the average rate of change of $f(x) = \sqrt{x+5}$ from 4 to x, then find its limit as $x \to 4$.

PCBC H Name **ANSWER KEY** **22** /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

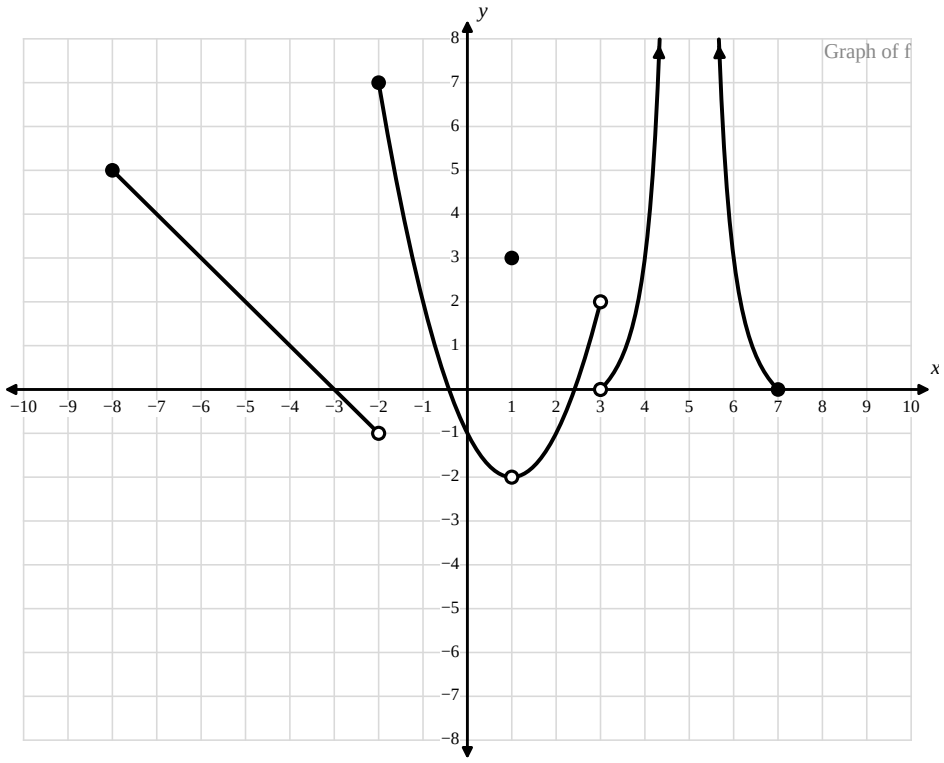
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SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)



1. $\lim_{x\to 1} f(x) =$

Both sides of the parabola head to the open circle at the vertex $(1, -2)$.

-2

2. $\lim_{x\to 3} f(x) =$

$\lim_{x\to 3^-} f(x) = 2$, $\lim_{x\to 3^+} f(x) = 0$

$2 \neq 0$, *so* $\lim_{x\to 3} f(x)$ *does not exist.*

DNE

3. $\lim_{x\to 5} f(x) =$

Both sides of $x = 5$ *rise without bound.*

∞

(2 points)

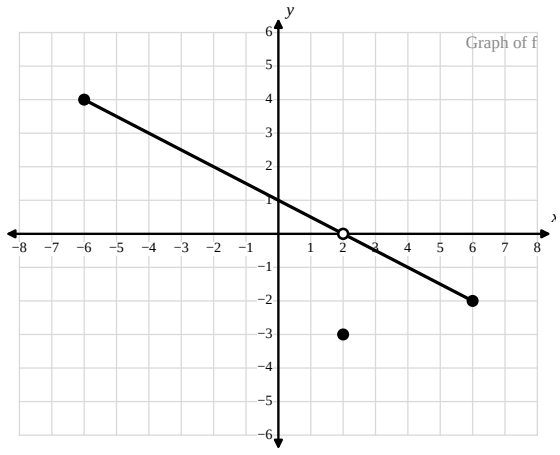
4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = -1$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

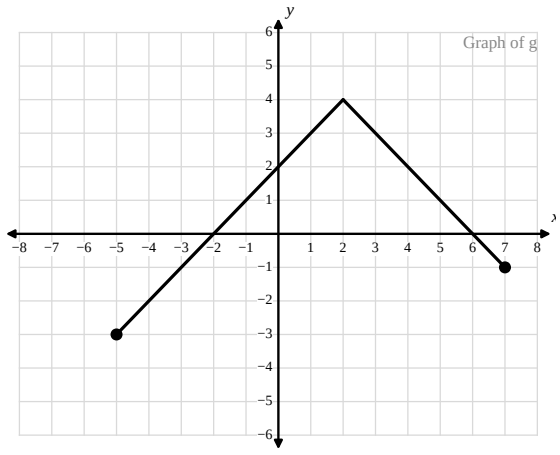
Height -1 *is reached from both sides at* $x = 0$ *and* $x = 2$.

Not $x = -2$*: only the left line heads to* -1*; the right side starts at 7.*

$a = 0,\ a = 2$

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim_{x\to 4}\sqrt{[g(x)]^2 - 3f(x)}$

$$\lim_{x\to 4}\sqrt{[g(x)]^2 - 3f(x)} = \sqrt{\left[\lim_{x\to 4} g(x)\right]^2 - 3\lim_{x\to 4} f(x)}$$

Root (inside limit is 7 > 0), power, difference, constant multiple.

$$= \sqrt{(2)^2 - 3(-1)} = \sqrt{4+3} = \sqrt{7}$$

$\sqrt{7}$

(2 points)

6. Use the table to estimate $\lim_{x\to 2} h(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	1.9	1.99	1.999	$\to 2 \leftarrow$	2.001	2.01	2.1
$h(x)$	11.41	11.9401	11.994001		12.006001	12.0601	12.61

Left: 11.41, 11.9401, 11.994001

Right: 12.61, 12.0601, 12.006001

Both sides agree. A table estimates; it does not prove.

≈ 12

$\lim_{x\to 2^-} h(x) \approx$ 12 $\lim_{x\to 2^+} h(x) \approx$ 12 $\lim_{x\to 2} h(x) \approx$ 12

Evaluate each limit. (2 points each)

7. $\lim_{x\to 1} f(x), \quad f(x) = \begin{cases} \frac{x^2-1}{x-1}, & x < 1 \\ 3 - x^2, & x \geq 1 \end{cases}$

$\lim_{x\to 1^-} \frac{x^2-1}{x-1} = \lim_{x\to 1^-} \frac{(x-1)(x+1)}{x-1} = \lim_{x\to 1^-} (x+1) = 2$

Left piece: substitution produces 0/0, so factor.

$\lim_{x\to 1^+} (3 - x^2) = 3 - 1 = 2$

$\lim_{x\to 1} f(x) = 2$

2

8. $\lim_{x\to \frac{1}{2}} \frac{2x^2 + 3x - 2}{4x^2 - 1}$

Substitution produces 0/0: indeterminate.

$\lim_{x\to \frac{1}{2}} \frac{(2x-1)(x+2)}{(2x-1)(2x+1)}$

$= \lim_{x\to \frac{1}{2}} \frac{x+2}{2x+1} = \frac{\frac{5}{2}}{2} = \frac{5}{4}$

$\frac{5}{4}$

9. $\lim_{x\to 3} \frac{x^2 - 3x}{\sqrt{x+1} - 2}$

Substitution produces 0/0: indeterminate.

$\lim_{x\to 3} \frac{x(x-3)\left(\sqrt{x+1}+2\right)}{(x+1)-4}$

$= \lim_{x\to 3} \frac{x(x-3)\left(\sqrt{x+1}+2\right)}{x-3}$

$= \lim_{x\to 3} x\left(\sqrt{x+1}+2\right) = 3(2+2) = 12$

12

10. $\lim_{x\to -1} \frac{\frac{1}{x+2} - 1}{x^2 - 1}$

Substitution produces 0/0: indeterminate.

LCD: x + 2

$\lim_{x\to -1} \frac{1 - (x+2)}{(x+2)(x^2-1)}$

$= \lim_{x\to -1} \frac{-(x+1)}{(x+2)(x-1)(x+1)}$

$= \lim_{x\to -1} \frac{-1}{(x+2)(x-1)} = \frac{-1}{(1)(-2)} = \frac{1}{2}$

$\frac{1}{2}$

11.

Find the average rate of change of $f(x) = \sqrt{x+5}$ from 4 to x, then find its limit as $x \to 4$.

$\frac{f(x) - f(4)}{x - 4} = \frac{\sqrt{x+5} - 3}{x - 4}$

Average rate of change; this is the first answer.

$\lim_{x\to 4} \frac{\sqrt{x+5} - 3}{x - 4} = \lim_{x\to 4} \frac{(x+5) - 9}{(x-4)\left(\sqrt{x+5}+3\right)}$

$= \lim_{x\to 4} \frac{1}{\sqrt{x+5}+3} = \frac{1}{3+3} = \frac{1}{6}$

$\frac{\sqrt{x+5}-3}{x-4}$; $\frac{1}{6}$